

Single-Photon Double Slit: Experiment III

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I. INTRODUCTION

The original double slit experiment performed by Thomas Young (1801) is one of the most well known experiments in the realm of optics and quantum mechanics, as it was one of the first experiments which demonstrated the *wave-like* nature of light.

The demonstration of *interference patterns* by Young contested the previously hypothesized “corpuscles” of light. Especially after the discovery of the photoelectric effect, which demonstrated *particle-like* behavior, the *wave-particle duality* of light was disputed for many years. This argument was fundamental to the development of quantum mechanics.

In particular, this experiment aims to demonstrate both wave-like and particle-like phenomena of light by performing the double slit experiment in such low intensity conditions that individual photons are present in the apparatus at a moment in time.

Expectations are such that individual *impulsive signals* can be detected and viewed with an oscilloscope such that a single “packet” of light—a photon—can be observed at one time. Each event demonstrates particle behavior. The spatial distribution of the locations of photon arrivals at a detector should reveal an interference pattern which is suggestive of wave behavior. This begs the question of how such interference occurs even when only one photon is present (See Ref. [1]).

II. THEORY

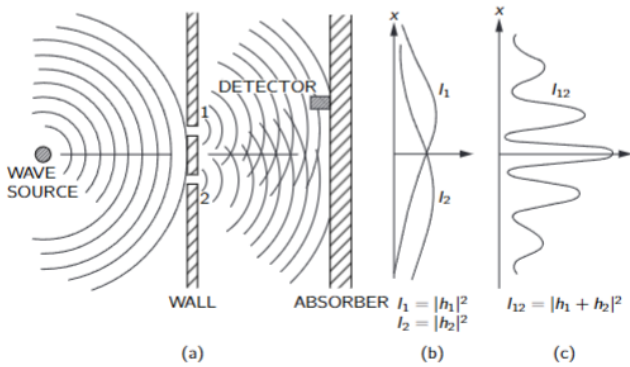


Fig. 1. See Ref. [2]. Illustration of how individual sources produce a diffraction pattern with constructive and destructive interference.

Because light waves interfere with each other, two separate sources will produce a diffraction pattern such that there are observable “bright” and “dark” regions. These regions occur where the waves from each source add *constructively* and *destructively*.

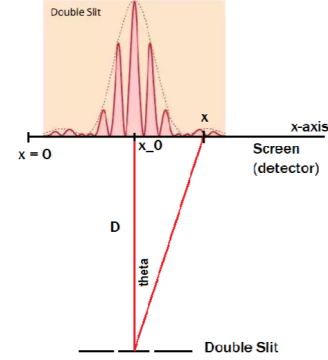


Fig. 2. Schematic diagram labeling angles and distances relating to the positions of the double slit and detector or screen, as well as a quantitative image illustrating the expected interference pattern. (See Ref. [3]).

The predictions of *Frauenhofer* theory produce a function of diffraction pattern intensity with respect to angular position, $I(\theta)$. For *single slit diffraction*,

$$I(\theta) = I(0) \cdot \left(\frac{\sin \alpha}{\alpha} \right)^2 \quad (1)$$

and in the case of *double slit diffraction*,

$$I(\theta) = 4I(0) \cdot \left(\frac{\sin \alpha}{\alpha} \right)^2 \cdot \cos^2 \beta \quad (2)$$

where $\alpha := \frac{ka}{2} \sin \theta$ and $\beta := \frac{kb}{2} \cos \theta$. The slit separation is roughly $b = \text{???}$, while the slit width is roughly $a = \text{???}$. The *propagation number* is defined as $k = 2\pi/\lambda$.

At a specified distance, D , from the double slit, the spatial position, x , of the detector is related to the angle, θ , by

$$\tan \theta = \frac{x - x_0}{D} \approx \sin \theta \quad (3)$$

for small angles. See Ref. [4] for derivations.

III. METHODS

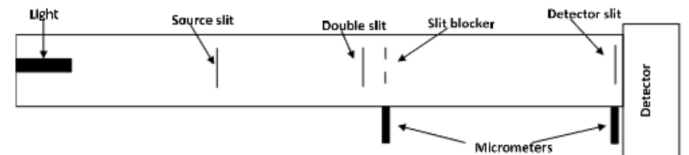


Fig. 3. Schematic diagram of the double-slit apparatus. The light source may either be a 5 mW laser or a low intensity lamp. The detector may be either a photodetector or a photomultiplier tube. A source slit is used to homogenize the light source, which is especially important when the lamp is in use.

The apparatus consists of a light source, from which light will propagate through a source slit in order to homogenize the source and ensure that plane waves are propagating through the instrument. From the source slit, light will propagate through a double slit, behind which a slit blocker can be used to block either slit at once. Finally, the light falls on a detector slit which will be used for selective measurement of the spatial distribution of light intensity. Behind the detector slit is either a *photodetector*, which produces a voltage that scales linearly with the intensity of the light absorbed, or a *photomultiplier tube* (PMT), which uses the photoelectric effect to generate significant voltage pulses from individual photon detection events.

Initially, some alignment is necessary in order to produce the diffraction pattern in the correct location such that it can be observed with the detector. Follow the steps highlighted in Ref. [5]. Additionally, the experimenter should note the micrometer values of five separate locations of the slit blocker such that the slit blocker can be operated while the apparatus is *closed* (See Ref. [1]):

- 1) Both slits are blocked.
- 2) Light emerges from the farther of two slits.
- 3) Light emerges from both slits.
- 4) Light emerges from the nearer of two slits.
- 5) Both slits are blocked (highest reading).

The experiment will be performed three different times, each with slight modifications. Initially, it will be performed qualitatively with a laser (670 ± 5 nm)—an index card will be placed to visually observe the interference pattern while the instrument enclosure is *open*. This should allow the experimenter to demonstrate the constructive and destructive addition of waves from both slits. Additionally, the slits which are open can be controlled to show that subtracting one light source from the system does in fact eliminate the fringe pattern. The pattern seen here is that of *Frauenhofer* diffraction.

The second portion of the experiment will be a repetition of the first part, but with a more quantitative approach that aims to perform accurate measurements of the pattern intensity and it's spatial distribution. These measurements are obtained by sliding the “detector slit” which permits light to pass through to the photodetector. This allows for a spatial mapping of the intensity distribution which can be compared to theoretical predictions of *Frauenhofer* diffraction.

The third and final sub-experiment will require a light bulb which can be set at such a low intensity that it will produce individual photons at one time. A filter is placed over the bulb such that photons are only produced in the small wavelength range of 541 to 551 nm. The PMT creates detectable voltage pulse signals from the arrival of individual photons through the diffracting apparatus. It will demonstrate that light does indeed arrive in finite, quantized packets as particles. Yet, the spatial distribution of these photons will still demonstrate the *Frauenhofer* diffraction predictions as far as spatial distribution of photon arrivals. (See Ref. [1] and [5] for full instructions on taking measurements).

IV. SAFETY CONSIDERATIONS

Both laser safety and electrical safety must be considered in this experiment.

Within the instrument is a 5 mW laser with the capability to cause eye damage only with direct exposure (i.e. looking directly into the focused beam). Since the laser is fixed inside the enclosed instrument, one must simply take care to avoid allowing the beam to be reflected out of the enclosure towards experiment participants. Jewelry and other reflective accessories should be removed when the instrument is in operation.

The *photomultiplier tube* (PMT) used to detect individual photons uses a powerful effect to produce a large voltage from only a single photon interaction, thus care must be taken to avoid exposing the PMT to even a dim light source. Exposure can damage or destroy the PMT and produce dangerously high voltages.

V. ANALYSIS & INTERPRETATION

In the second experiment, we will aim to produce a plot of intensity (voltage with some scale factor) against the position of the detector slit. This way we can “map” the spatial distribution of the diffraction pattern quantitatively. The analysis will require some sort of fitting to the *Frauenhofer* diffraction equations, and should be compared with expected results based on the parameters of the apparatus (laser light frequency, slit separation, slit width, etc.). In fact, the fit of measurements should be used with error propagation to get calculate the slit width and separation and their uncertainty. The actual scaling factor between the voltage measured and the intensity is just some arbitrary constant, so it's not so important to get exactly the right amplitude, just an equivalent shape, then the scale factor can be added in to correct the units and match the predictions of theory. The plot will look something like the Desmos plotting of the diffraction pattern in the experiment information library.

The third part of the experiment will require a slightly different approach. Individual photons must be counted with the oscilloscope. Photons will arrive at some low background rate in the “dark” areas of the fringe pattern, and at some rate on the order of 10^3 per second in the “bright” regions. We will be recording the photon counts at one or more 1-second intervals for various positions of the detector slit when both slits are open. A similar plot will be generated as in part two, but instead the dependent variable will be an actual photon count, rather than a voltage corresponding to an intensity. We will repeat this analysis for three cases: one slit, the other slit, and both slits open.

It will be important to establish a “background rate” of photons for which both slits are blocked, which will appear as a vertical offset in our plots.

The ultimate question to remain unanswered here is... if there is only one photon present in the double slit apparatus at one time, what could it possibly be interfering with?

VI. CONCLUSIONS

The main conclusions of this paper are simply to demonstrate the ability of the apparatus to produce results which

agree with the predictions of the Fraunhofer diffraction theory, in conjunction with the necessary fit parameters of the apparatus and light sources. The results will likely also show the need for theory which does not make the same simplifying assumptions as Fraunhofer diffraction does.

The point of the experiment is not to answer the question of wave-particle duality, but rather to demonstrate why terms such as *duality* needed to be invented. The results of this experiment are the very mystery of quantum mechanics.

REFERENCES

- [1] TeachSpin, *2-Slit Interference, 1 Photon at a Time: Operating Manual*, 2003.
- [2] R. Feynman, *The Feynman Lectures on Physics, Volume III*. Pearson, 1964.
- [3] “2-Slit Interference, 1 Photon at a Time: Fitting Hints,” *CarmenCanvas*, 2025.
- [4] E. Hecht, *Optics*. Pearson, 2016.
- [5] “2-Slit Interference, 1 Photon at a Time v2.3,” *CarmenCanvas*, 2022.